

PROFESSIONAL EXPERIENCE

IBM Research HQ, NY | Research Engineer

May 2025 - Aug 2025

- Built a production-ready agentic DataOps platform with CrewAI, MCP tools, and ReAct planning, that automates data-product creation from NL using question generation, Text-to-SQL, dynamic views [\[link\]](#).
- Developed a metric driven orchestration agent that reasons over data quality metrics and autonomously selects tools to improve schema coverage, query efficiency, and product usability.
- Engineered a human and agent operable dashboard that manages multi-agent conversations and workflows, leveraging asyncio and task queues for concurrency and distributed communication.
- Built a plug-and-play integration layer supporting PostgreSQL, SQLite, BigQuery containerized with Docker.

Dealsplus Pvt. Ltd, UK | SDE - III

Dec 2022 - July 2024

- Developed graph model to represent complex private equity ownership data, increasing rendering speed from 1.8s to 0.5s (-72%) at 10k+ nodes/edges, driving a 40% client acquisition boost, adding \$1M in revenue.
- Led implementation of automated ownership data extraction using Azure Document Intelligence, 24 pages/min with precision/recall 92%/89%, reducing manual review time by 40%.
- Optimized database queries, increasing data retrieval efficiency by 38%, enabling faster, more accurate cap table outputs for improved decision-making.
- Designed the org's testing framework (unit + E2E), automating critical workflows (deploy time 20 to 8 mins) to minimize regressions (< 3% rollback) and accelerate development cycles.

Avinya Technologies, Bengaluru | Co-Founder & Senior Tech Member

April 2021 - Dec 2022

- Directed software solutions development, enabling a 70% expansion in retail outlets over two years with only a 25% workforce increase.
- Created ML algorithms to predict retail demand, reducing overstocking and working capital needs by 35%.
- Built the POS system and Spring Boot back end, deployed across ~50 stores reducing checkout time 42% with an offline-first, reliable sync model.
- Designed scalable accounting solutions automating payment reconciliation, handled WhatsApp/email notifications out of the box cutting manual back-office time by ~60%.

D.E. Shaw & Co, Hyderabad | Member Tech - Risk Tech Team

Dec 2018 - Mar 2021

- Collaborated and maintained the ETL pipelines and the risk service that powers daily and month-end positions across multi-strategy portfolios, partnered with quants/risk to ship safely under close deadlines.
- Engineered for peak traffic and reworked hot paths and caching for time-series data analysis by collaborating with cache management team and increasing cache hit by 88% and reducing backend calls by 41%.
- Turned ad-hoc analyses into repeatable pipelines, shipping comparative workflows for ~600k+ monthly/FX positions per close across 1,200+ instruments and built a rule-based risk data monitoring engine that surfaced anomalies, reducing manual triage and making risk analyst reviews smoother.
- Consolidated tools into a single risk portal and modernized the stack from Py2 to Py3, increased test coverage by 16%, improved p95 page-load by ~1 min, and documented run-books for smooth on-call handoffs.

RESEARCH & PROJECT EXPERIENCE

Agentic AI DataOps ([IBM Research](#)): Automating Semantic Data Product Evaluation and Optimization via Metric-Driven Decision Systems using AI Agents. First author of the research demo paper accepted in ICDM 2025.

Dexterous Manipulation ([PAIR Lab](#)): Researched dexterous manipulation using Reinforcement Learning with [Prof. Animesh Garg](#). Main contributor to survey paper comparing Deep Learning techniques [\[link\]](#).

RAG: Evaluated locally runnable LLMs for textbook QA and built a RAG pipeline evaluating chunking strategies, BM25/TF-IDF retrieval, golden-set evals, and reranking for strong source grounding guided by [Prof. Joy Arulraj](#).

GPT-2 Implementation: Implemented GPT-2 from scratch using Python and PyTorch, replicating its architecture, including multi-head attention, positional encodings, and feed-forward layers [\[Github\]](#)

Clinical Co-Pilot: Built an AI-driven healthcare co-pilot integrating Gemini, Claude, and GPT-4 for clinical workflows, including transcription, ICD/CPT code extraction, and medical image summarization and automatic insurance filing [\[demo\]](#).

EDUCATION

M.S. in CS, Georgia Institute of Technology, Atlanta, Georgia

Aug 2024 - Dec 2025

B.E. in CS, PSG College of Technology, Coimbatore, India

May 2015 - June 2019

SKILLS

Languages: Python, Java, C, C++, C#, JavaScript, Typescript, HTML, CSS, SQL, Go, Rust, Bash

Frameworks & Tools: React, Next.js, Spring Boot, Hibernate, Android, Unity, YOLO, CUDA, PyTorch, TensorFlow, Keras, Hugging Face, LangChain, Crew AI, MLflow, MLOps, AIOps, IsaacLab, IsaacSim, Docker, Kubernetes, Jenkins, Nginx, Azure Cloud, Firebase, Git, Selenium, Celery, Splunk, Grafana, Redis, Jest, Mocha, Chai, Haystack, Pinecone, Weaviate, Milvus, Chroma, FAISS, Neo4j, D3, GoJS, Ragas, DeepEval, Pydantic